Qns	Answer	Marks
8(a)1	Since the circular path is in the clockwise direction it suggests that the magnetic force must point rightwards (from left to right) Using Fleming's Left Hand Rule, (when the magnetic field is out of the paper, and the magnetic force is towards the right), this is only possible if the particle is positively charged.	B1 B1
8(a)2	For the positively charged particle to accelerate upwards it must experience an electric force that acts in the upward direction. This occurs only if the electric field points upwards which corresponds to a higher electric potential at the bottom plate.	B1 B1
8(b)(i)	The electric field between parallel plates is uniform. $$E = \frac{\Delta V}{\Delta d} = \frac{V_0}{d} = \frac{V}{0.5d}$$ Hence, $V = \frac{V_0}{2}$	M1 A0
8(b)(ii)	Gain in kinetic energy = Loss in electric potential energy $\frac{1}{2}mv^2 - 0 = q\Delta V = \frac{qV_0}{2}$ $\therefore v^2 = \frac{qV_0}{m}$ $v = \sqrt{\frac{qV_0}{m}}$	M1 A0
8(c)	The magnetic force always acts perpendicular to the direction of velocity (infinitesimal tangential displacement along the charged particle's path). No work is done, its kinetic energy remains constant and thus, the speed does not increase.	B1 B1

Qns	Answer	Marks
8(d)	The magnetic force (F_B) provides for the centripetal force (F_c) $F_B = F_c$ $Bqv = mr\omega^2 \Rightarrow Bq(r\omega) = mr\omega^2$ $\therefore \omega = \frac{Bq}{m}$ $\omega = \frac{2\pi}{T}$ $\frac{2\pi}{T} = \frac{Bq}{m}$ $\therefore T = \frac{2\pi m}{Bq}$ Since the particle travels half a circle, the time travelled in the upper dee is: $t = \frac{T}{2}$ $= \frac{\pi m}{Bq}$ **Alternative solution:** The magnetic force (F_B) provides for the centripetal force (F_c) $F_B = F_c$ $Bqv = \frac{mv^2}{r} \Rightarrow v = \frac{Bpr}{m}$ Since $v = \frac{\pi r}{t}$, where πr is the distance travelled and t is the time taken for the semicircular path $\frac{\pi r}{t} = \frac{Bqr}{m}$ $\therefore t = \frac{\pi m}{Bq}$	B1 C1 M1 M1 A0 B1 C1 M1 M1 A0
8(e)	The time taken is independent of the velocity of the particle (only depends on m, B and q), Hence, it will always be the same as that found in (d).	M1 A1
8(f)	t = 1.6 ns $B = \frac{\pi m}{tq} = \frac{\pi}{(1.6\times10^{-9})(1.0\times10^{8})} = 19.63495 = 20$ T (2.s.f)	C1 A1
8(g)	At the upper dee, the charged particle has a kinetic energy of $\frac{qV_0}{2}$ Gain in kinetic energy = Loss in electric potential energy = qV_0	 C1

Qns	Answer	Marks
	Thus, total kinetic energy = $\frac{qV_0}{2} + qV_0 = 1.5qV_0$	A1
8(h)	When the particle first travels into the upper dee, it has a kinetic energy of 0.5 qV_0. The total kinetic energy of the particle can then be written as E_K = (0.5 + n) qV_0, where n is the number of times the particle is in either of the dees after exiting the upper dee for the first time. Since the kinetic energy is 6.5 qV_0, n = 6. By doing a manual count, the particle will exit from the upper dee and the it passes the lower dee is a total of 3 times. **#Food for thought:** To generalize this: If n = odd, then we can tell that the particle exits from the lower dee and the number of times it will pass the lower dee is $\frac{(n+1)}{2}$ If n = even, then we can tell that the particle exits from the upper dee and the number of times it will pass the lower dee is $\frac{n}{2}$	C1 A1